Space Debris Detection System

Phase 1 Comprehensive Engineering Report & Verification Record

ISRO Problem Statement #05 | Phase 01: Foundations & Day-1 Audit | Date: October 16, 2026

✓ 100% VERIFIED & SIGNED OFF

5 / 5 UAT GATES PASSED

Executive Summary: Phase 1 successfully established the rigorous foundations of the Edge-Deployable Space Debris Detection System. We implemented the decoupled evaluation harness (`src/eval.py`), completed the Day-1 SPARK CSV label audit across 80,000 bounding boxes (locking DEC-04 Branch B), and verified real-time inference on edge compute with **85.94 ms latency (11.6 FPS)** and **~207 MB peak memory**, well within the ≤ 100 ms and ≤ 512 MB constraints.

DETECTION ACCURACY

99.50%

mAP@0.5 (94.13% mAP@0.5:0.95, 99.96% Precision)

EDGE LATENCY (4-THREAD)

85.94 ms

11.6 FPS (≤ 100 ms real-time edge target passed)

DATASET AUDIT

80,000

Labels audited across 10 classes (0 debris rows, DEC-04)

1. Dual-Workstation Matrix & Collaboration Architecture

Development and validation were executed collaboratively across two hardware platforms, clearly separating the responsibilities of the **ML Lead** and the **Data & Edge Lead**:

Dimension	Compute Server: NVIDIA Quadro GV100	Edge Testbed: Laptop / Embedded Pi 4
Assigned Role	**ML Lead** (Model Architecture & Deep Training)	**Data & Edge Lead** (Ingestion, Audit & Edge Benchmarks)
Processor	High-Core Workstation CPU (16 Cores)	Intel Core i5-11400H / ARM Cortex-A72 Quad Core
GPU Accelerator	NVIDIA Quadro GV100 (32 GB HBM2 VRAM)	NVIDIA GeForce RTX 2050 (4 GB GDDR6)
Primary Responsibility	50-Epoch YOLOv8-Nano Training, Export, GPU mAP	CSV Audits, Photometric Augmentation, 4-Thread Benchmarks
Software Stack	CUDA 12.1, PyTorch 2.5, Ultralytics YOLOv8	Python 3.10/3.11, ONNX Runtime 1.31, NCNN, psutil

2. Standalone Evaluation Harness (`src/eval.py`)

Prior to Phase 1, the repository had no independent evaluation CLI (`src/eval.py` was an empty stub). Model metrics could only be observed inside the training loop. We implemented a decoupled evaluation CLI adhering to the EVAL-02 metric contract.

```
------------------------------------------------
  STANDALONE EVALUATION SUMMARY (SPARK 10-Class)
------------------------------------------------
  mAP@0.5:      0.9950  (99.50%)
  mAP@0.5:0.95: 0.9413  (94.13%)
  Precision:    0.9996  (99.96%)
  Recall:       0.9986  (99.86%)
  GPU Latency:  18.50 ms (RTX 2050) / 3.44 ms (GV100)
  Model Size:   6.29 MB (best.pt)
------------------------------------------------
```

3. Day-1 SPARK Dataset Annotation Audit & Decision DEC-04

The official SPARK challenge literature referenced an 11th class for generic `debris`. On Day 1, we executed `scripts/audit_spark_annotations.py` to inspect all ground-truth labels and prevent training on phantom annotations.

Class Name	Train Count	Validation Count	Total Count	Debris Flag	Status
Cheops	6,000	2,000	8,000	No	**Verified**
LisaPathfinder	6,000	2,000	8,000	No	**Verified**
ObservationSat1	6,000	2,000	8,000	No	**Verified**
Proba2	6,000	2,000	8,000	No	**Verified**
Proba3	6,000	2,000	8,000	No	**Verified**
Proba3ocs	6,000	2,000	8,000	No	**Verified**
Smart1	6,000	2,000	8,000	No	**Verified**
Soho	6,000	2,000	8,000	No	**Verified**
VenusExpress	6,000	2,000	8,000	No	**Verified**
XMM Newton	6,000	2,000	8,000	No	**Verified**
TOTALS	**60,000**	**20,000**	**80,000**	**0 (0.00%)**	**DEC-04 Lock**

> **Architectural Decision DEC-04:** Confirmed 80,000 labels across 10 spacecraft classes and 0 debris rows. Locked **Branch B**: train the primary multi-class detector on the 10 real spacecraft classes, maintaining SwissCube and SPEED+ as testbed/occlusion data.

4. Rigorous Edge Inference Benchmarking (`edge/benchmark.py`)

We refactored `edge/benchmark.py` to follow research-grade benchmarking rigor: ≥10 warmup runs discarded, ≥100 timed runs, thread pinning (1 and 4 threads matching Raspberry Pi 4 cores), and peak process Resident Set Size (RSS) profiling via `psutil`.

Engine & Backend	Threads / Device	Input Size	Mean Latency	p95 Latency	FPS	Peak RSS Memory	Budget Status
ONNX Runtime (CPU)	1 Thread (Single Core)	1024x1024	182.38 ms	191.49 ms	5.5	204.7 MB	Baseline
ONNX Runtime (CPU)	**4 Threads (Pi 4 Simulated)**	**1024x1024**	**85.94 ms**	**97.62 ms**	**11.6**	**207.1 MB**	**PASS (≤ 100 ms)**
PyTorch / CUDA	Local RTX 2050 GPU	1024x1024	18.50 ms	21.10 ms	54.1	~650 MB	Dev Machine
PyTorch / CUDA	Quadro GV100 GPU	1024x1024	3.44 ms	4.10 ms	290.7	~1.2 GB	Compute Server

5. Phase 1 Verification Sign-Off Matrix (5/5 PASSED)

Test #	Feature Subsystem	Acceptance Criteria & Observed Output	UAT Status
Test 1	Evaluation Harness (`src/eval.py`)	Independent metric calculation on GPU/CPU: mAP@50 = 99.50%, mAP@50-95 = 94.13%, Latency = 18.5 ms.	✓ **PASSED**
Test 2	SPARK CSV Audit (`audit_spark_annotations.py`)	80,000 bounding boxes audited across 10 classes, 0 debris instances, DEC-04 Branch B confirmed.	✓ **PASSED**
Test 3	Edge Environment (`edge/verify_pi_env.py`)	Python 3.11, onnxruntime 1.31, ncnn, psutil, CPU core & RAM verified.	✓ **PASSED**
Test 4	Edge Benchmark (`edge/benchmark.py`)	10 warmup, 100 runs: 85.94 ms latency (≤ 100 ms), 11.6 FPS, ~207 MB RSS (≤ 512 MB).	✓ **PASSED**
Test 5	Repo Hygiene & Configs	Decoupled YAML configs (`spark.yaml`, `swisscube.yaml`), `requirements-edge.txt`, `.gitignore` active.	✓ **PASSED**

6. Phase 2 Transition Roadmap

With all Phase 1 foundations locked, the team advances directly into **Phase 2: SPARK Dataset Preparation & Stratified Subsetting**:

- **Subsetting Strategy:** Standardize images to 640x640 resolution (DEC-06) and subsample majority trajectories to 2:1 ratio for balanced training (~24k train, 3k val, 3k test).
- **Leakage Prevention:** Enforce trajectory-aware splitting so that all frames from a single sequence remain exclusively in train or validation.
- **Photometric Augmentation:** Apply train-split-only transforms via `src/data/augment.py` (4,030 augmented baseline images already validated).
- **Training Readiness:** Deliver clean processed dataset to the ML Lead for Phase 4 overnight training on the Quadro GV100 and RTX 2050.
